

RITIKA BIST

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EDUCATION

M.Sc. Bioinformatics

2025 – 2027

Banasthali Vidyapith

B.Sc. Life Sciences

2021 – 2024

J.C. Bose University of Science and Technology, YMCA

RELEVANT COURSEWORK

Completed: Molecular Biology, Genomics, Biostatistics, Database Management, Bioinformatics Algorithms

In Progress / Planned: Machine Learning for Bioinformatics, Structural Bioinformatics

EXPERIENCE

Bioinformatics Research Trainee ICAR-IASRI, New Delhi

May 2026 – Jun 2026

- Developed modular shell-scripted pipelines for end-to-end NGS data processing, integrating QC, alignment, and quantification stages into reproducible workflows executable on HPC clusters
- Performed comparative transcriptomic analysis across biological conditions, evaluating differential expression patterns and interpreting results in the context of underlying biological mechanisms
- Worked within a CSIR-funded research environment alongside PhD-level researchers, contributing to active genomics projects on agricultural crop systems
- Attended PhD-level GWAS lectures, building working knowledge of genome-wide association study methodology and statistical genetics

PROJECTS

Transcriptomic Analysis of Therapy-Induced Senescence & PQQ Intervention

2026

Independent research project | Dataset: GSE290476 (NCBI GEO) | Human dermal fibroblasts

- Built a complete 7-phase bulk RNA-seq pipeline from raw FASTQ files to biologically interpreted differential expression results — independently, without institutional supervision
- Pipeline: FastQC → MultiQC → Trimmomatic → HISAT2 (GRCh38, 91–92% mapping rate) → StringTie → DESeq2 across 9 samples (85,321 genes)
- Identified 7,954 DEGs under bleomycin-induced senescence and 2,579 DEGs under PQQ senotherapeutic intervention; findings consistent with SASP activation, DNA damage response, and cell cycle arrest pathways
- Generated full visualization suite in R: PCA plots, volcano plots, MA plots, and hierarchical heatmaps; PC1 captured 38% variance with clear condition-specific clustering

<https://github.com/sikkibist/transcriptomic-analysis-senescence>

TECHNICAL SKILLS

Bioinformatics: NGS analysis, Differential expression, Pipeline development

Tools: FastQC, MultiQC, Trimmomatic, HISAT2, STAR, StringTie, DESeq2, BLAST, OMICSBOX

Programming: Python, R, Perl, Shell scripting

Systems: Linux/Unix, HPC environments, MySQL, XML, HTML